



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Name: Aashray Lal Kayastha

PRN: 25070122005

Branch: CSE (2025-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No.: 4

TITLE: :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement:

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives:

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory:

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Outcomes:

Wrapper classes in Java (Integer, Double, Character, Boolean, etc.) allow primitive data types to be treated as objects. This experiment demonstrated how to convert primitive values into their corresponding wrapper objects (boxing) and back into primitives (unboxing), how to convert String input into usable numeric types using methods like `parseInt()` and `parseDouble()`, and how wrapper class methods can be used to perform comparisons and basic operations on wrapped values — concepts essential for working with Java Collections and handling real-world user input.

Conclusion:

This experiment demonstrated the use of wrapper classes to convert between primitive data types and objects. Methods like `parseInt()` and `parseDouble()` converted `String` input into usable numbers, while `compareTo()` enabled validation, showing how wrapper classes support real-world data handling in Java.

Exercise:

1. Create a program to convert student marks from `String` format to `Integer` and calculate total marks.

```
class Student{  
  
    String name;  
    String mark1Str, mark2Str, mark3Str;  
  
    Student(String n, String m1, String m2, String m3){  
  
        name = n;  
        mark1Str = m1;  
        mark2Str = m2;  
        mark3Str = m3;  
    }  
  
    void calculateTotal(){  
        int mark1 = Integer.parseInt(mark1Str);  
        int mark2 = Integer.parseInt(mark2Str);  
        int mark3 = Integer.parseInt(mark3Str);  
  
        int total = mark1 + mark2 + mark3;  
    }  
}
```

```

        double average = total/3.0;

        System.out.println("Name: " + name);
        System.out.println("Marks: " + mark1 + ", " + mark2 + ", " + mark3);
        System.out.println("Total: " + total);
        System.out.println("Average: " + average);
    }

    public class JAVA_EXP_3{
        public static void main(String[] args){
            Student s1 = new Student("Abel", "75", "80", "90");
            s1.calculateTotal();
        }
    }
}

```

Output:

```

C:\Users\Aashray\OneDrive\Desktop\JAVA_PRACTICALS>java JAVA_EXP_3.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Name: Aashray
Marks: 75, 80, 90
Total: 245
Average: 81.66666666666667

```

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```

class Employee{
    String empId;
    Double basicSalary;
    Double bonus;

    Employee(String id, String salaryStr, String bonusStr){
        empId = id;
        basicSalary = Double.parseDouble(salaryStr);
        bonus = Double.parseDouble(bonusStr);
    }

    void calculateNetSalary(){

        if(basicSalary.compareTo(0.0) <= 0){
            System.out.println("Invalid basic salary for Employee ID: " + empId);
            return;
        }
    }
}

```

```

    }
    if(bonus.compareTo(0.0) < 0){
        System.out.println("Invalid bonus amount for Employee ID: " + empId);
        return;
    }

    double netSalary = basicSalary + bonus;

    System.out.println("Employee ID: " + empId);
    System.out.println("Basic Salary: " + basicSalary);
    System.out.println("Bonus: " + bonus);
    System.out.println("Net Salary: " + netSalary);
}
}

public class JAVA_EXP_3{
    public static void main(String[] args){
        Employee e1 = new Employee("E101", "45000", "5000");
        e1.calculateNetSalary();

        System.out.println();

        Employee e2 = new Employee("E102", "-2000", "1000");
        e2.calculateNetSalary();
    }
}

```

Output:

```

C:\Users\Aashray\OneDrive\Desktop\JAVA_PRACTICALS>java JAVA_EXP_4.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Employee ID: E101
Basic Salary: 45000.0
Bonus: 5000.0
Net Salary: 50000.0

Invalid basic salary for Employee ID: E102

```